

## Lecture-01

# 0612-302 Data Communication

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# CHAPTER-1

## *Introduction*

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*Textbook:* Data Communications and Networking  
Fourth Edition

- Behrouz A. Forouzan

# Overview

- Data communication.
- Characteristics of data communication.
- Components of data communication.
- Data representation.
- Data flow.
- Network.
- Criteria of a network.
- Type of connection.
- Physical topology.
- Categories of network.
- Protocol.

# Data Communications

- **Data communications** are the exchange of data between two devices via some form of transmission medium such as a wire cable.
- For data communications to occur, the communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs).
- The effectiveness of a data communications system depends on four fundamental characteristics: delivery, accuracy, timeliness, and jitter.

# Data Communications

- **Delivery:** The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user.
- **Accuracy:** The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.
- **Timeliness:** The system must deliver data in a timely manner. Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called real-time transmission.
- **Jitter:** Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets. For example, let us assume that video packets are sent every 3 ms. If some of the packets arrive with 3 ms delay and others with 4 ms delay, an uneven quality in the video is the result.

# Components

- A data communications system has five components (see Figure 1.1).

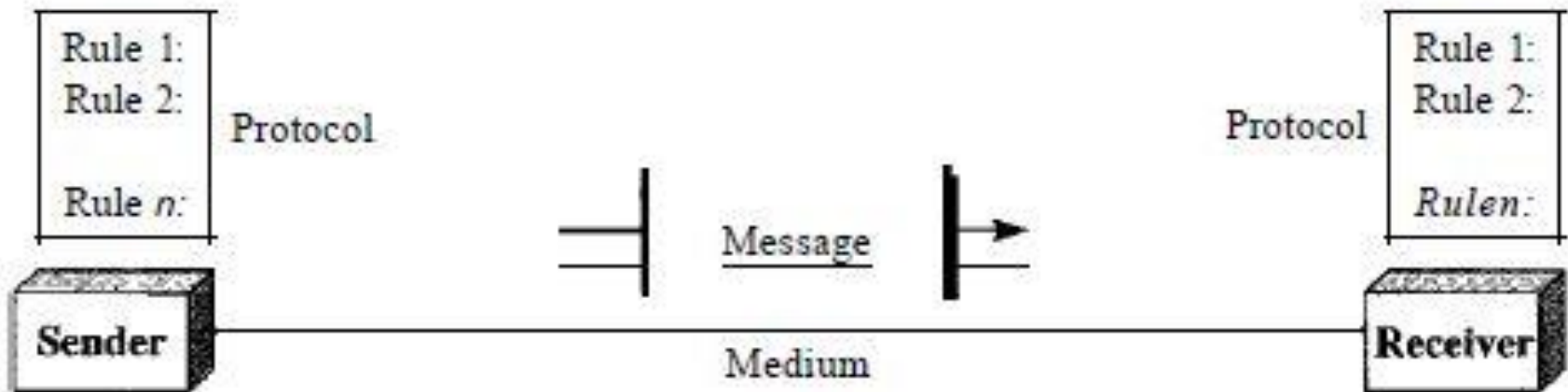


Figure 1.1: Five components of data communication

# Components (contd.)

1. **Message:** The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.
2. **Sender:** The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.
3. **Receiver:** The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.
4. **Transmission medium:** The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.
5. **Protocol:** A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

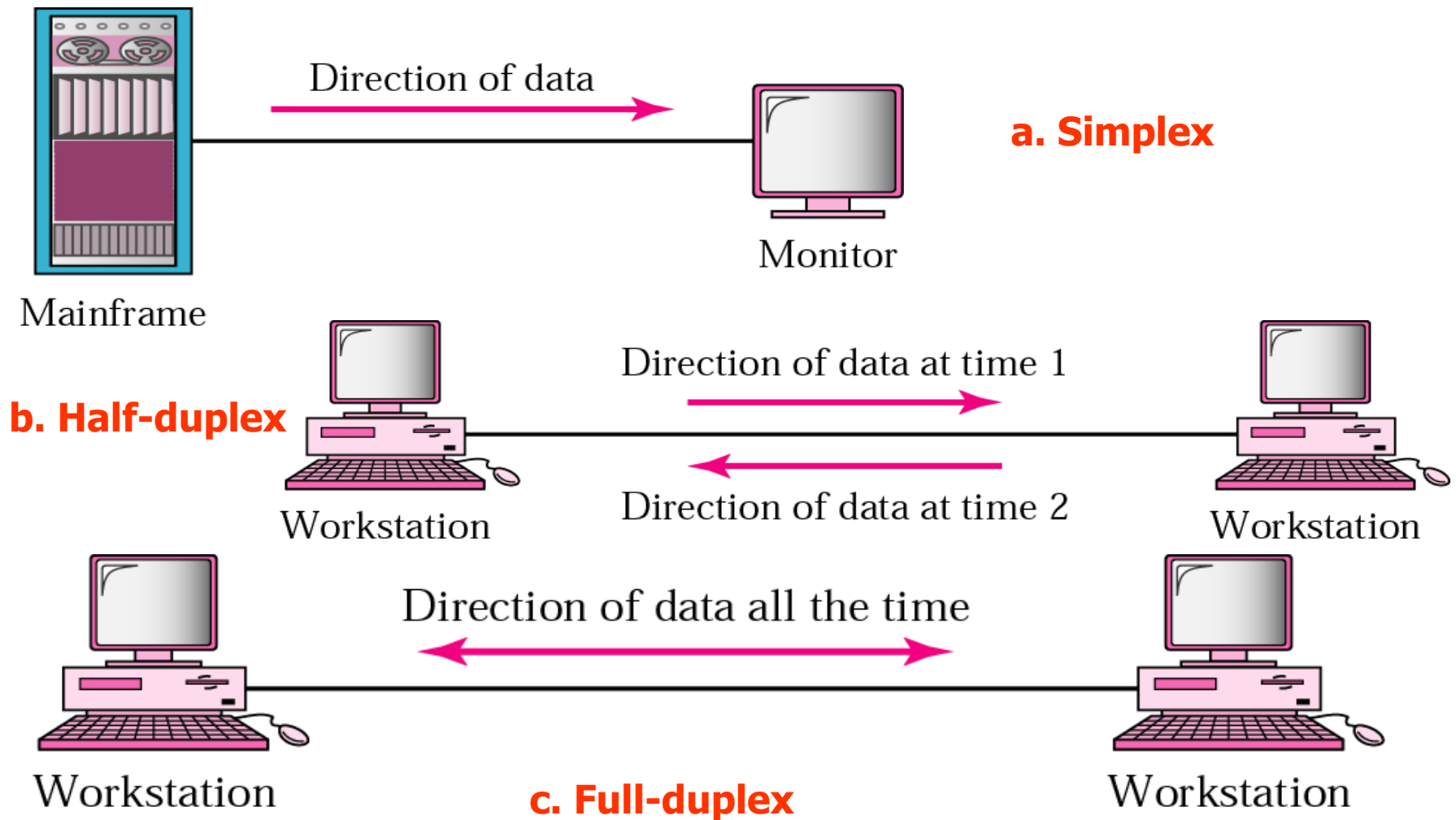
# Data Representation

- Text
  - Represented as a bit pattern
  - Various codes: ASCII (7 bits), Extended ASCII (8 bits), Unicode (16 bits), ISO (32 bits)
- Numbers
  - Directly converted to a binary number
- Images
  - jpeg, gif, etc.
- Audio
  - mp3, wma,
- Video
  - mp4, avi, etc.

# Data Flow

- Communication between two devices can be simplex, half-duplex, or full-duplex as shown in Figure 1.2.
- **Simplex:** In simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive (see Figure 1.2a).
- Keyboards and traditional monitors are examples of simplex devices. The keyboard can only introduce input; the monitor can only accept output.
- The simplex mode can use the entire capacity of the channel to send data in one direction.

# Direction of Data Flow



**Figure 1.2: Data flow (simplex, half-duplex, and full-duplex)**

# Data Flow

- **Half-Duplex:** In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa (see Figure 1.2b).
- The half-duplex mode is like a one-lane road with traffic allowed in both directions. When cars are traveling in one direction, cars going the other way must wait.
- In a half-duplex transmission, the entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time.
- Walkie-talkies and CB (citizens band) radios are both half-duplex systems.

# Data Flow

- **Full-Duplex:** In full-duplex mode (also called duplex), both stations can transmit and receive simultaneously (see Figure 1.2c).
- The full-duplex mode is like a two-way street with traffic flowing in both directions at the same time.
- In full-duplex mode, signals going in one direction share the capacity of the link: with signals going in the other direction.
- This sharing can occur in two ways: Either the link must contain two physically separate transmission paths, one
- for sending and the other for receiving; or the capacity of the channel is divided between signals traveling in both directions.
- One common example of full-duplex communication is the telephone network. When two people are communicating by a telephone line, both can talk and listen at the same time.

# Networks

- **Network** : A network is a set of devices (often referred to as nodes) connected by communication links.
- A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.
- **Network Criteria**: A network must be able to meet a certain number of criteria. The most important of these are-
  - Performance,
  - Reliability, and
  - Security.

## *Criteria of a Network*

- **Performance:** Can be measured in many ways, including transit time and response time.
  - Transit time is the amount of time required for a message to travel from one device to another device.
  - Response time is the elapsed time between an inquiry and a response.
- The performance of a network depends on a number of factors, including the number of users, the type of transmission medium, the capabilities of the connected hardware, and the efficiency of the software.
- Performance is often evaluated by two networking metrics: throughput and delay. We often need more throughput and less delay.

## *Criteria of a Network*

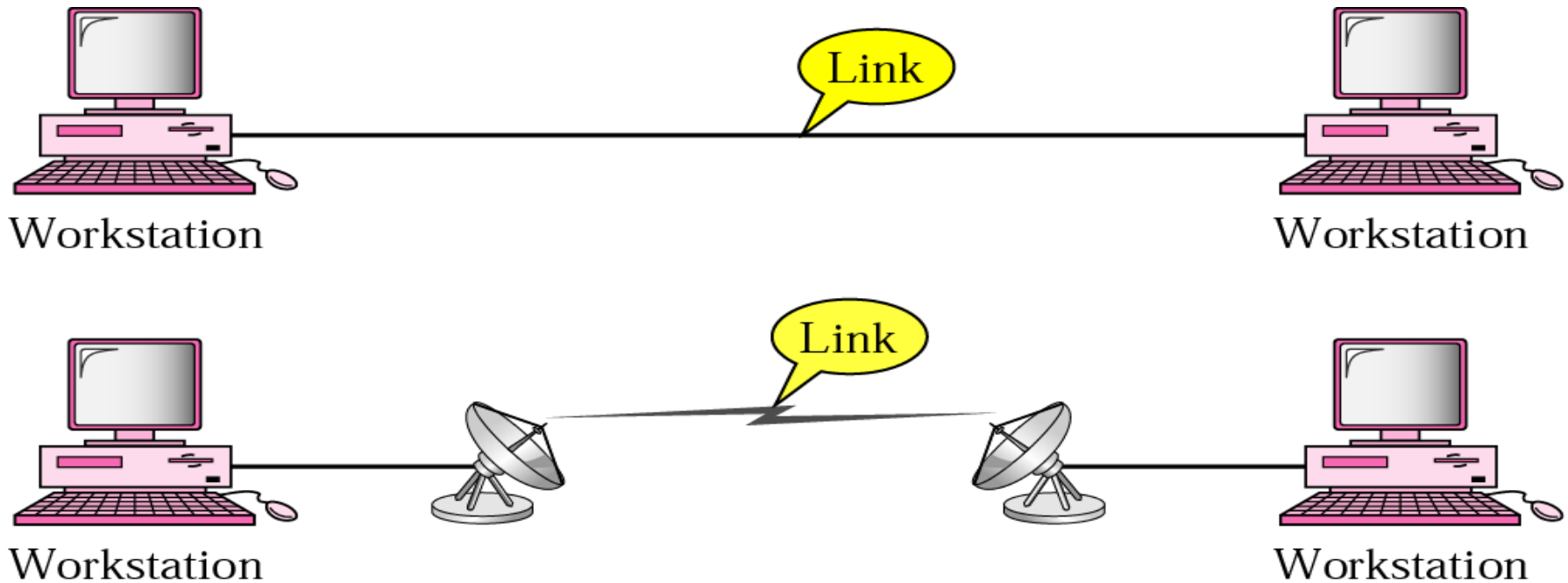
- However, these two criteria are often contradictory. If we try to send more data to the network, we may increase throughput but we increase the delay because of traffic congestion in the network.
- **Reliability:** In addition to accuracy of delivery, network reliability is measured by the frequency of failure, the time it takes a link to recover from a failure, and the network's robustness in a catastrophe.
- **Security:** Network security issues include protecting data from unauthorized access, protecting data from damage and development, and implementing policies and procedures for recovery from breaches and data losses.

# Type of Connection

- A network is two or more devices connected through links. A link is a communications pathway that transfers data from one device to another.
- For visualization purposes, it is simplest to imagine any link as a line drawn between two points.
- For communication to occur, two devices must be connected in some way to the same link at the same time.
- There are two possible types of connections:
  - Point-to-point and
  - Multipoint.

# Point-to-Point Connection

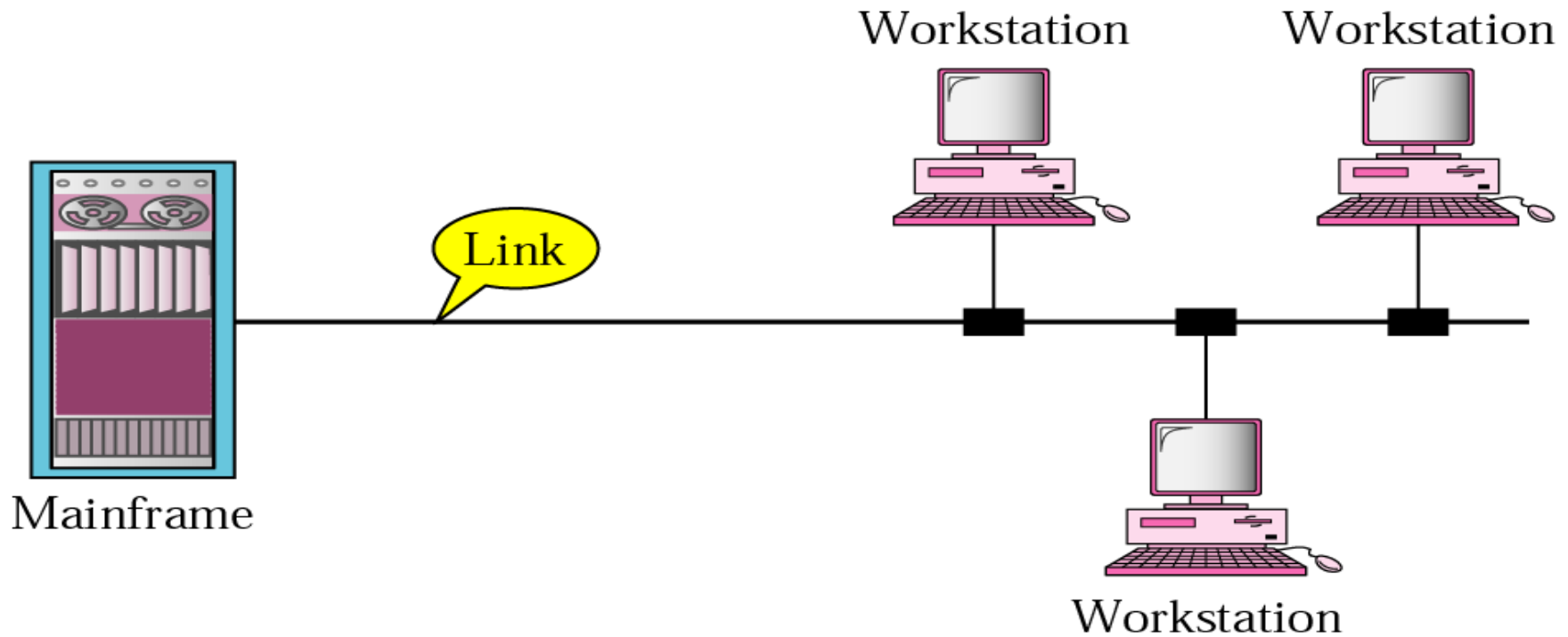
A **point-to-point** line configuration provides a dedicated link between two devices. The entire capacity of the channel is reserved for transmission between those two devices. When you change television channels by infrared remote control, you are establishing a point-to-point connection between the remote control and the television's control system.



**Figure 1.3: Types of connections: point-to-point**

# Multi-point Connection

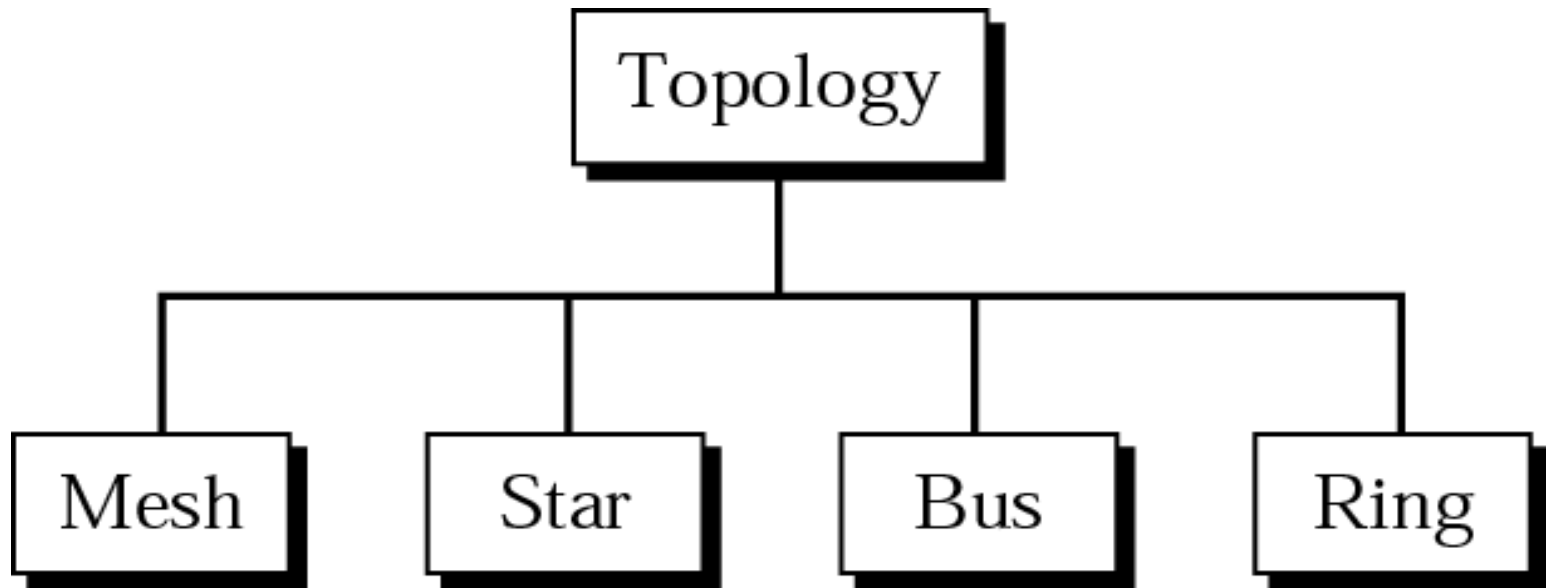
**Multipoint** (also called multidrop) connection is one in which more than two specific devices share a single link. In a multipoint environment, the capacity of the channel is shared, either spatially or temporally.



**Figure 1.3: Types of connections: multipoint**

# Physical Topology

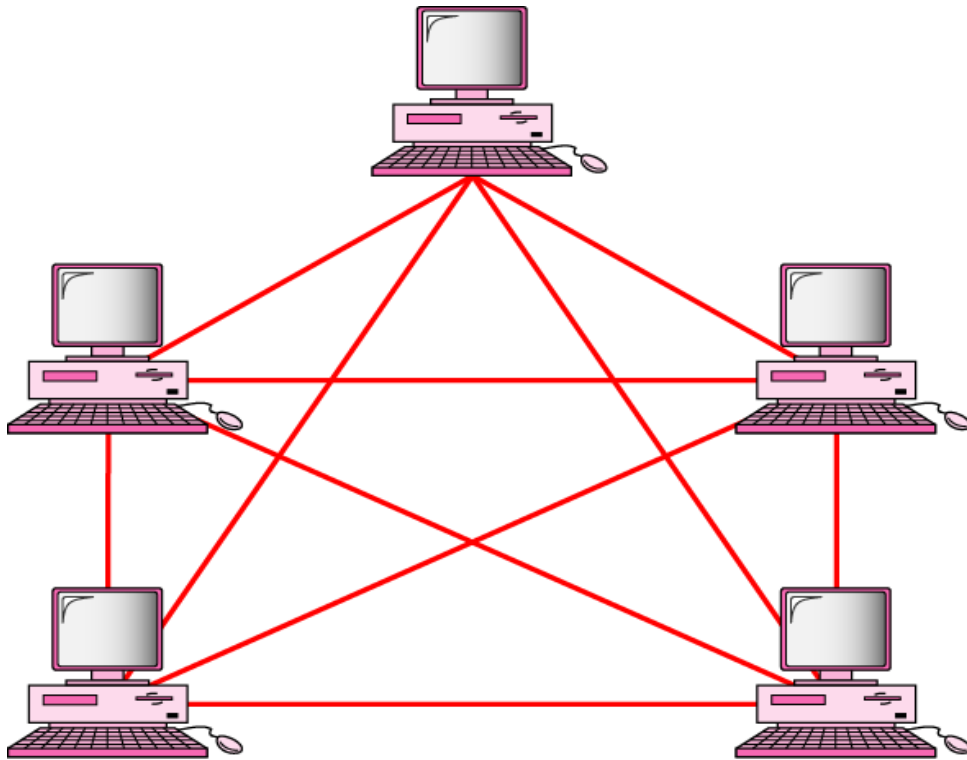
The term physical topology refers to the way in which a network is laid out physically. Two or more devices connect to a link; two or more links form a topology. There are four basic topologies possible: mesh, star, bus, and ring.



**Figure 1.4:** *Categories of topology*

# Mesh Topology

In a **mesh topology**, every device has a dedicated point-to-point link to every other device. The term dedicated means that the link carries traffic only between the two devices it connects.



**Figure 1.5: A fully connected mesh topology (five devices)**

# Mesh Topology

- To find the number of physical links in a fully connected mesh network with  $n$  nodes, we first consider that each node must be connected to every other node.
- Node 1 must be connected to  $n - 1$  nodes, node 2 must be connected to  $n - 1$  nodes, and finally node  $n$  must be connected to  $n - 1$  nodes. We need  $n(n - 1)$  physical links.
- However, if each physical link allows communication in both directions (duplex mode), we can divide the number of links by 2.
- In other words, we can say that in a mesh topology, no of duplex-mode links =  $n(n - 1) / 2$ .

# Mesh Topology (contd.)

## Advantages

- The use of a dedicated links guarantees that each connection can carry its own load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.
- A mesh topology is robust. If one link becomes unusable, it does not incapacitate the entire system.
- Privacy: When every message sent travels along a dedicated link, only the intended recipient sees it. Physical boundaries prevent other users from gaining access to messages.
- Point-to Point links make fault identification and fault isolation easy.

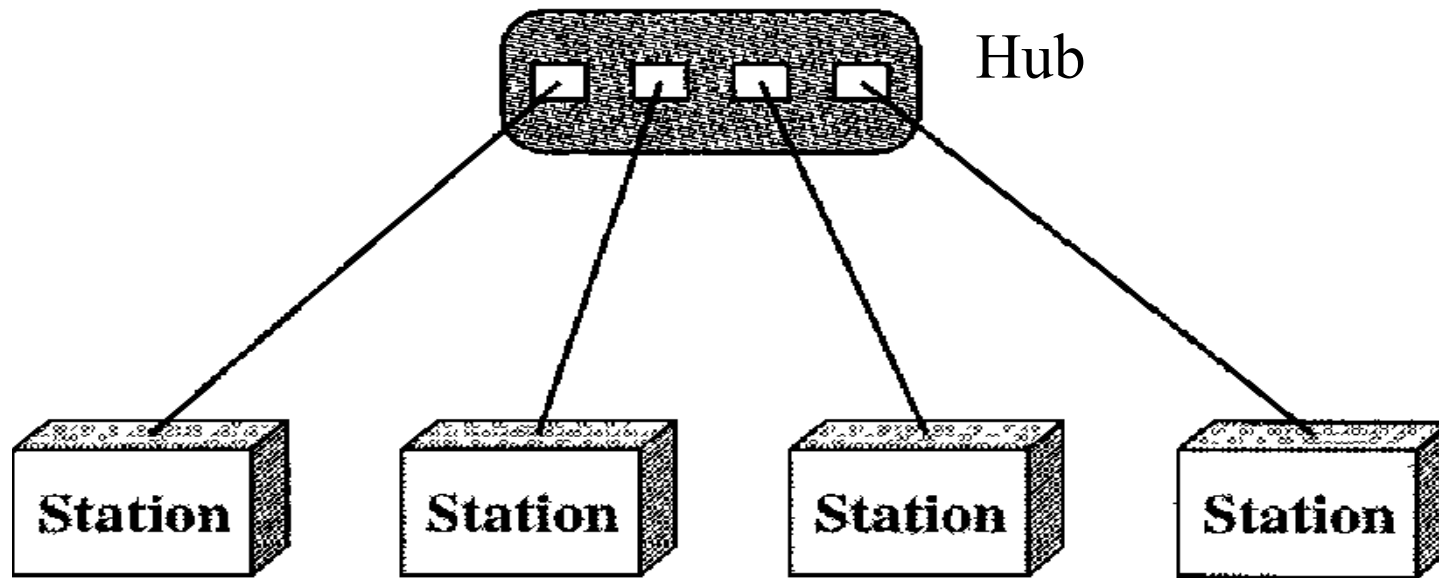
# Mesh Topology (contd.)

## Disadvantages

- A big amount of cabling and a big number of I/O ports required, i.e., the hardware required can be very expensive.
- Because every device must be connected to every other device, installation and reconfiguration are difficult.
- For these reasons a mesh topology is usually implemented in a limited fashion.

# Star Topology

In a **star topology**, each device has a dedicated point-to-point link only to a central controller, usually called a hub (or switch). A star topology does not allow direct traffic between devices. If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device.



**Figure 1.6: A star topology connecting four stations**

# Star Topology (contd.)

## Advantages

- It is easy to install and reconfigure.
- The star topology is robust. If one link fails, only that link is affected. This factor also lends itself to easy fault identification and fault isolation.

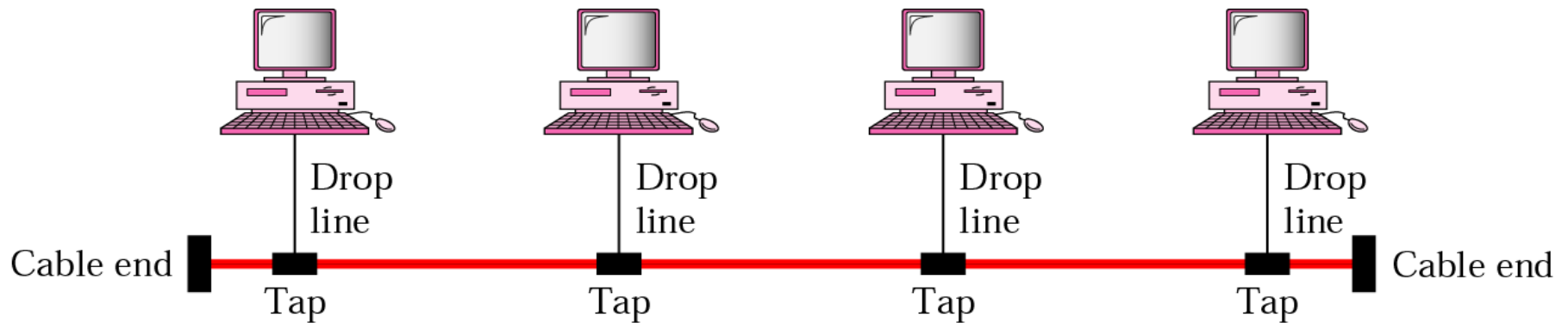
## Disadvantages

- If the central hub goes down, the whole system is dead.
- Although a star requires far less cable than a mesh, each node must be linked to a central hub. For this reason, often more cabling is required in a star than in some other topologies (such as ring or bus).

**Note:** The star topology is used in local-area networks (LANs). High-speed LANs often use a star topology with a central hub.

# Bus Topology

A **bus topology**, on the other hand, is multipoint. One long cable acts as a backbone to link all the devices in a network. Nodes are connected to the bus cable by drop lines and taps.



**Figure 1.7: A bus topology connecting three stations**

# Bus Topology

- A drop line is a connection running between the device and the main cable.
- A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.
- As a signal travels along the backbone, some of its energy is transformed into heat.
- Therefore, it becomes weaker and weaker as it travels farther and farther.
- For this reason there is a limit on the number of taps a bus can support and on the distance between those taps.

# Bus Topology (contd.)

## Advantages

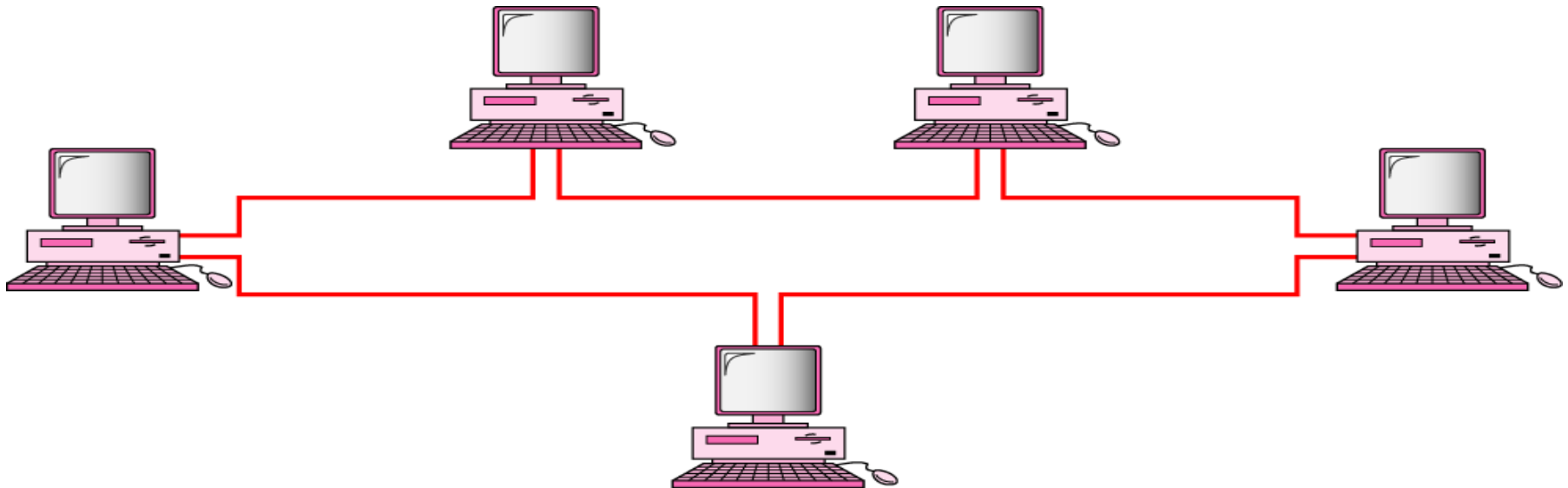
- Easy to install.
- Bus topology uses less cabling than mesh or star topology.

## Disadvantages

- Difficult reconfiguration and fault isolation. A bus is usually designed to be optimally efficient at installation. It can therefore be difficult to add a new devices.
- A fault or break in the bus cable stops all transmission, even between devices on the same side of the problem. The damaged area reflects signals back in the direction of origin, creating noise in both directions.
- Bus topology was the one of the first topologies used in the design of early local area networks.
- Ethernet LANs can use a bus topology, but they are less popular now

# Ring Topology

In a **ring topology**, each device has a dedicated point-to-point connection with only the two devices on either side of it. A signal is passed along the ring in one direction, from device to device, until it reaches its destination. Each device in the ring incorporates a repeater. When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along.



**Figure 1.8: A ring topology connecting six stations**

# Ring Topology (contd.)

## Advantages

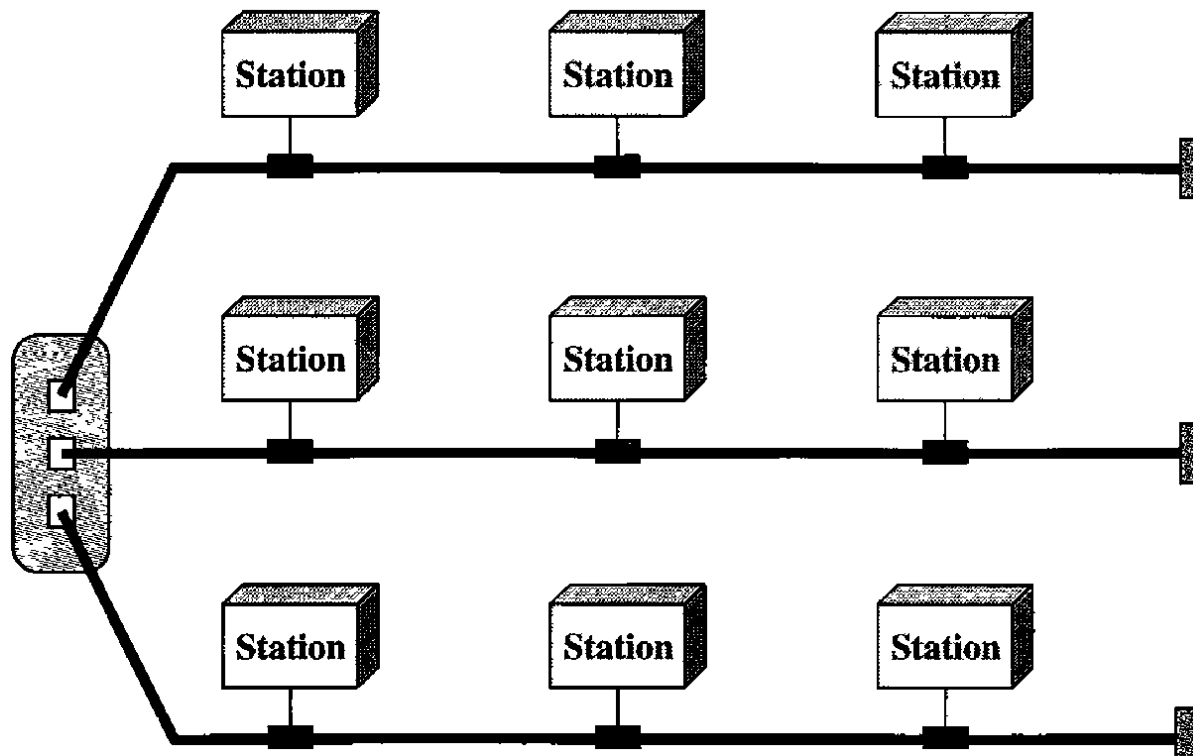
- Easy to install and reconfigure. To add or delete a device require only two connections.
- Fault isolation is simplified. Generally in a ring, a signal is circulating at all times. If one device does not receive a signal within a specified period, it can issue an alarm. The alarm alerts the network operator to the problem and its location.

## Disadvantages

- A break in the ring (such as a disabled station) can disable the entire network.
- This weakness can be solved by using a dual ring or a switch capable of closing off the break.
- Note: Ring topology was prevalent when IBM introduced its local-area network Token Ring. Today, the need for higher-speed LANs has made this topology less popular.

# Hybrid Topology

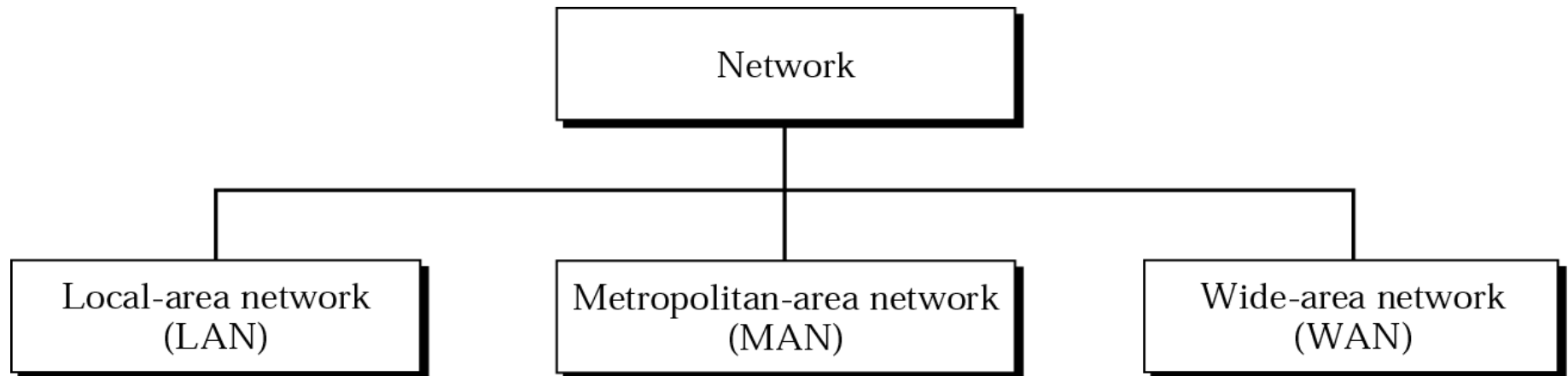
- A network can be hybrid.
- For example, we can have a main star topology with each branch connecting several stations in a bus topology.



**Figure 1.9: A hybrid topology: a star backbone with three bus networks.**

# Categories of Networks

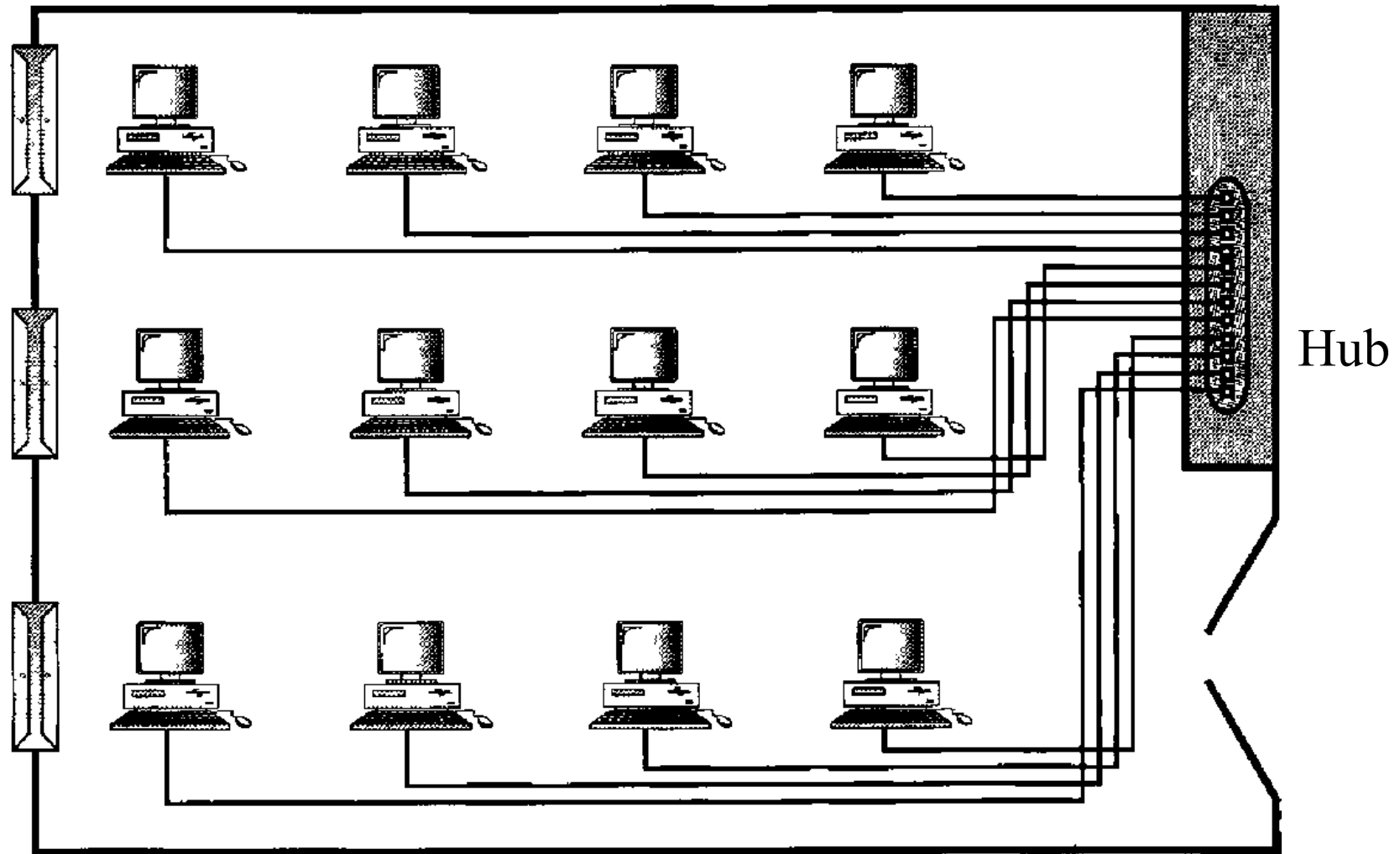
- Today when we speak of networks, we are generally referring to two primary categories: **local-area networks** and **wide-area networks**.
- The category into which a network falls is determined by its size.
- A LAN normally covers an area less than 2 mi; a WAN can be worldwide.
- Networks of a size in between are normally referred to as metropolitan area networks and span tens of miles.



# Local Area Networks (LAN)

- A **local area network (LAN)** is usually privately owned and links the devices in a single office, building, or campus.
- Depending on the needs of an organization and the type of technology used, a LAN can be as simple as two PCs and a printer in someone's home office; or it can extend throughout a company and include audio and video peripherals.
- Currently, LAN size is limited to a few kilometers.
- In general, a given LAN will use only one type of transmission medium. The most common LAN topologies are bus, ring, and star.
- Early LANs had data rates in the 4 to 16 megabits per second (Mbps) range. Today, however, speeds are normally 100 or 1000 Mbps.
- Wireless LANs are the newest evolution in LAN technology.

# Local Area Networks (LAN)

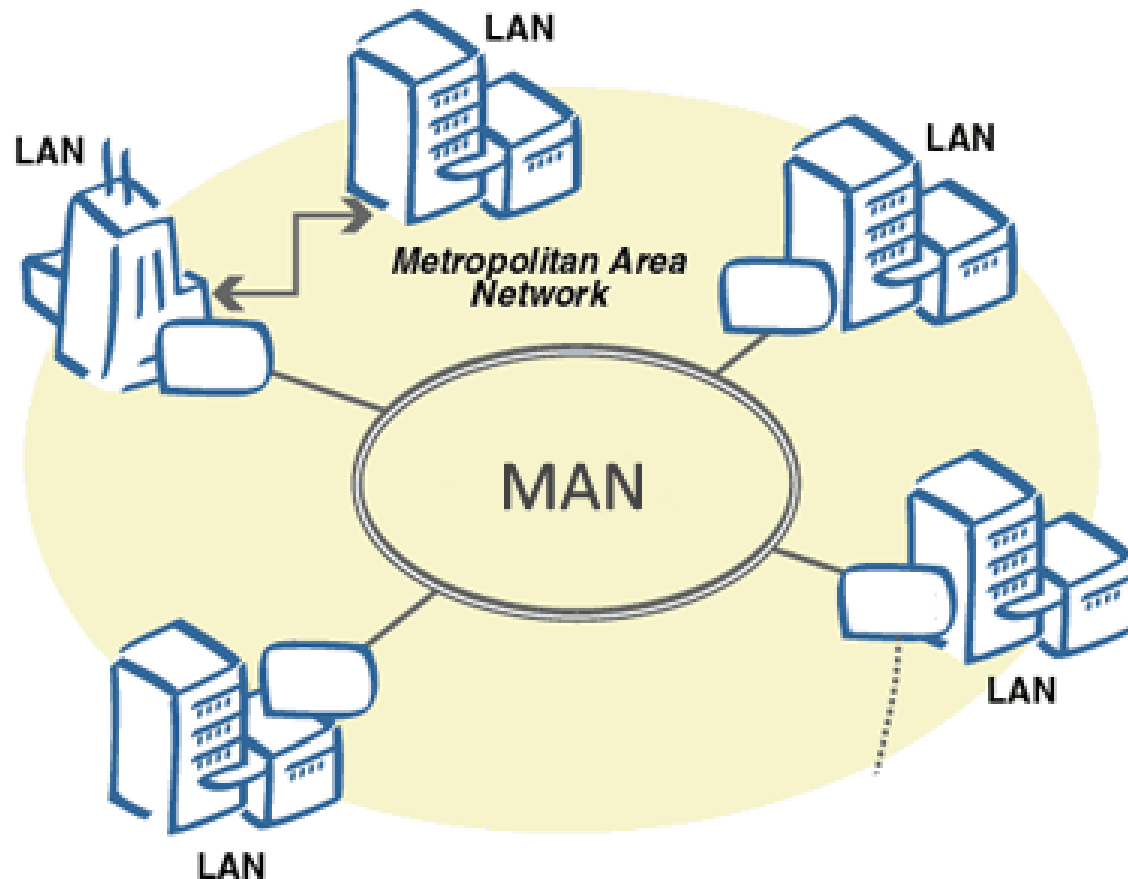


**Figure 1.10: An isolated LAN connecting 12 computers to a hub in a closet**

# Metropolitan Area Network

- A **metropolitan area network (MAN)** is a network with a size between a LAN and a WAN. It normally covers the area inside a town or a city.
- It is designed for customers who need a high-speed connectivity, normally to the Internet, and have endpoints spread over a city or part of city.
- A good example of a MAN is the part of the telephone company network that can provide a high-speed DSL line to the customer.
- Another example is the cable TV network that originally was designed for cable TV, but today can also be used for high-speed data connection to the Internet.

# Metropolitan Area Networks (MAN)

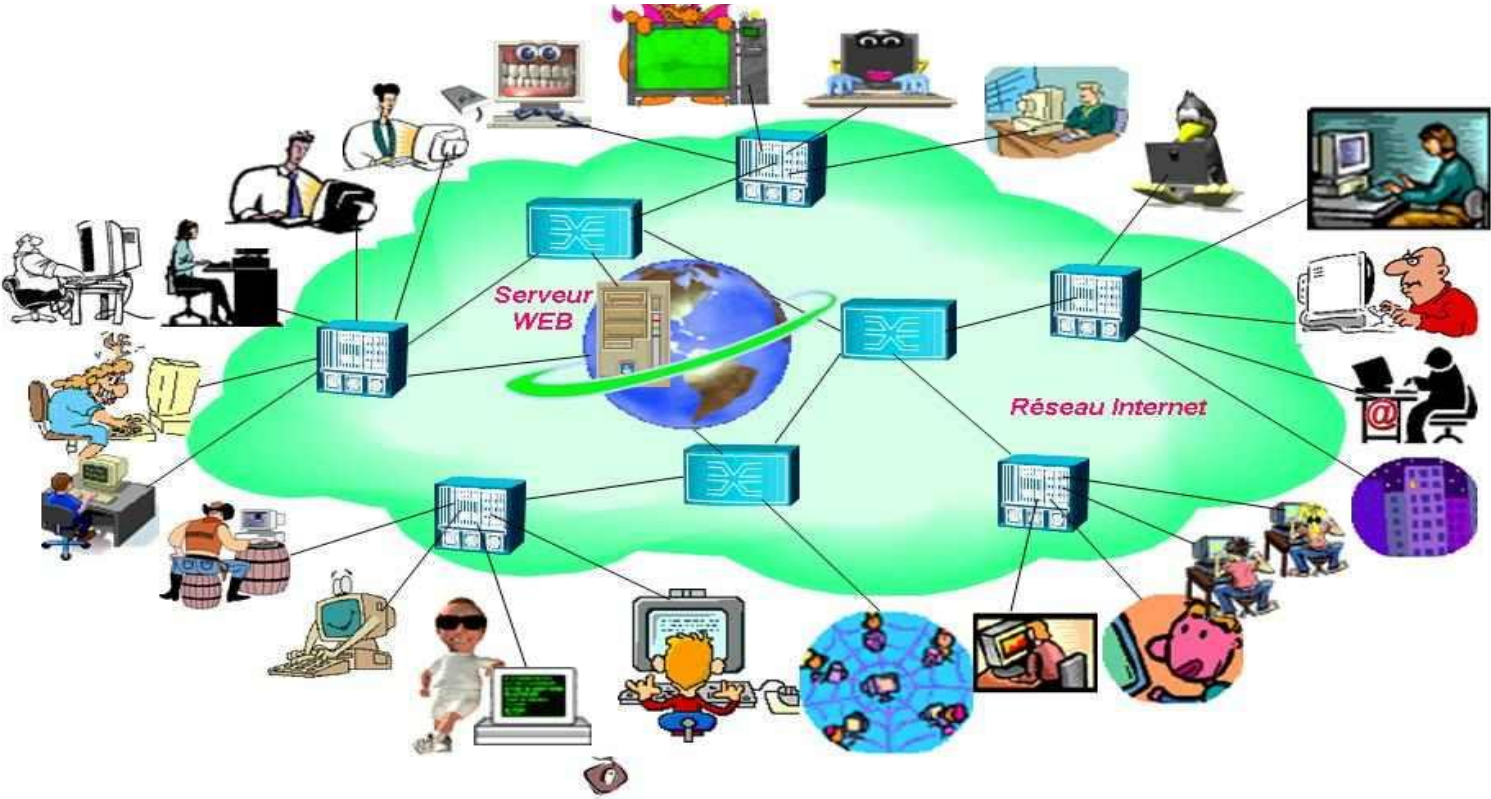


**Figure: Metropolitan Area Network (MAN)**

# Wide Area Networks (WAN)

- A wide area network (WAN) provides long-distance transmission of data, image, audio and video information over large geographic areas that may comprise a country, a continent, or even the whole world.
- The main difference between a MAN and a WAN is that the WAN uses Long Distance Carriers.
- Otherwise the same protocols and equipment are used as a MAN.

# Wide Area Networks



Internaute : Mondial 1 milliard – France 25 millions – UK 40 millions.

### Figure 1.11: Wide Area Network (WAN)

# Wide Area Network

- A **Wide Area Network** is a network spanning a large geographical area of around several hundred miles to across the globe.
- May be privately owned or leased.
- Also called “enterprise networks” if they are privately owned by a large company.
- Can be connected through cable, fiber or satellite.
- Is typically slower and less reliable than a LAN.
- Services include internet, Intranet ,etc.

# LAN, MAN, WAN



**Figure: LAN, MAN, WAN**

# DIFFERENCE BETWEEN LAN, WAN AND MAN

| BASIS            | LAN                                                                                          | MAN                                                                                      | WAN                                                                                      |
|------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Full Form        | Local Area Network                                                                           | Metropolitan Area Network                                                                | Wide Area Network                                                                        |
| Range            | A communication network linking a number of stations in same local area. Range is 1 to 10 km | This network shares the characteristics of packet broadcasting networks. Range is 100 km | A communication network distinguished from a Local Area Network. Range is Beyond 100 km  |
| Media Used       | Uses guided media                                                                            | Uses guided as well as unguided media                                                    | Uses unguided media                                                                      |
| Speed            | A high speed i.e. 100kbps to 100mbps                                                         | Optimized for a large geographical area than LAN.                                        | Long distance communications, which may or may not be provided by public packet network. |
| Cost             | cheaper                                                                                      | costly                                                                                   | expensive                                                                                |
| Equipment needed | NIC, switch and hub                                                                          | Modem and router                                                                         | Microwave, radio, infra-red laser                                                        |
| protocols        | Attached Resource computer network (ARCNET), Token ring                                      | Frame relay and asynchronous transfer mode(ATM)                                          | ATM, FDDI, SMDS                                                                          |

# Protocols

- In computer networks, communication occurs between entities in different systems.
- An entity is anything capable of sending or receiving information.
- However, two entities cannot simply send bit streams to each other and expect to be understood.
- For communication to occur, the entities must agree on a protocol.
- A **protocol** is a set of rules that govern data communications.
- A protocol defines what is communicated, how it is communicated, and when it is communicated.
- The key elements of a protocol are:- syntax, semantics and timing.

# Protocols

- **Syntax:** The term syntax refers to the structure or format of the data, meaning the order in which they are presented.
- For example, a simple protocol might expect the first 8 bits of data to be the address of the sender, the second 8 bits to be the address of the receiver, and the rest of the stream to be the message itself.
- **Semantics:** The word semantics refers to the meaning of each section of bits. How is a particular pattern to be interpreted, and what action is to be taken based on that interpretation?
- For example, does an address identify the route to be taken or the final destination of the message?
- **Timing:** The term timing refers to two characteristics: when data should be sent and how fast they can be sent.
- For example, if a sender produces data at 100 Mbps but the receiver can process data at only 1 Mbps, the transmission will overload the receiver and some data will be lost.

# Questions???

- Define data communication.
- Write the characteristics of data communication.
- Describe the components of data communication.
- Classify data communication based on the direction of data flow.
- State computer network.
- Illustrate the criteria of a computer network.
- Describe different types of connections of data communication.
- Describe different topologies with figure, advantages and disadvantages.
- Demonstrate LAN, MAN and WAN.
- Differentiate among LAN, MAN and WAN.
- Describe protocol with key elements.



THANKS  
EVERYBODY